



NETWORK DESIGN TEAM

(325)

PRELIMINARY

NATIONAL 2022

Description

Analyze existing and planned business environments and develop a strategy for the implementation of a network infrastructure that addresses the business needs of the scenario provided.

Topic

The world of esports is a massively connected multi-billion-dollar industry. DF Solutions, a technology support contractor for small, medium, and large-scale conference/special events, has recently decided to enter the world of esports event planning. Your team's task is to create a network infrastructure plan (replicable models) to accommodate Regional, State, and National esports conferences based on the information below.

- Local or Regional esports Conference:
 - This model should define the requirements for creating a small-scale network to accommodate up to 200 players.
- State esports Conference:
 - This model should define the requirements for creating a mid-scale network to accommodate up to 1000 players.
- National esports Conference:
 - This model should define the requirements for creating a large-scale network to accommodate up to 5000 players.

Requirements to consider for all levels:

- Bandwidth requirements to accommodate low-latency gameplay
- Secure storage (network shared) for all games
- Anti-cheat systems
- Closed vs open network design
- Equipment requirements (computers, routers, switches, etc.)

Teams who do *not* submit an entry following this topic will be *disqualified*.

Modification for National Preliminary:

Latency and packet loss are the biggest concerns when running an esports event. How do your models propose solutions to both of these challenges?